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M-407 CHLORINATION PLANT

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Subsection 1.2. "Summary of work".

The new Chlorination Plant shall be provided for Block 2, 3 & 4 circulating water system with approximately flow **80 m³/h**. The Chlorination Plant shall consist of two (2) trains for combined cycle Blocks 2, 3 and 4 with one (1) trains for normal operation and one (1) for stand by. ADD1 -IIC- 002

Each train of Chlorination Plant shall be complete with accessories , including but shall not be limited to the following :

- Two (2) x 100% duty sodium hypochlorite generator (electrolytic cells)
- Two (2) x 100% duty transformer and rectifier.
- Two (2) x 100% duty seawater pump with double mechanical seals and fresh water flush.
- Two (2) x 100% duty self cleaning seawater strainer and/or pretreatment equipment.
- Diffusers.
- Two (2) x 100% duty chlorination injection pump with double mechanical seals and fresh water flush.
- Two (2) residual chlorine sampling pump.
- One (1) residual chlorine analyzer test skid portable type.
- Two (2) Sodium hypochlorite hydrogen removal tank with level controller to modulate an air operated flow control valve downstream of injection pumps.
- All necessary pipe work, valves, fittings and appurtenances for the complete system including.
- One (1) system control panel. (for two trains)

- All necessary instrumentation, controls, pump drive motors, transformer/rectifiers and miscellaneous electrical equipment required for the safe and reliable operation of the plant.
- Air conditioning system for control panel and transformer cooling.
- Building with structural steel, foundations, lighting and access doors.
- Ventilation equipment for the Building.
- Personnel safety equipment.

The Chlorination Injection System for block 2 shall be connected from the existing Chlorination Plant. The Bidder shall provide one (1) new Sodium Hypochlorite tank connected with existing Chlorination Plant and 2 x 100% Chlorine Injection Pumps with all necessary pipe work, valves, fittings and appurtenances for the complete system including.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5. The following codes and standards extracted from Subsection 2.5 “Codes and Standards” or other equivalent international standards are specifically applicable to the design, manufacture and testing of the equipment in this Specification.

ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for testing and Material
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association

3. DESIGN AND OPERATING CONDITIONS

This Electrochlorination System shall be designed to be operated and controlled locally.

The injections for the electrochlorination are before bar screen for the circulating water system.

Consideration shall be given to design of the equipment in view of preventing pollution.

This system shall be designed to permit easy operation and maintenance.

- (a) The operating conditions of pumps and any other equipment shall be controlled and monitored from the control panel.
- (b) Should there be any possibility of trouble in operation and control due to trouble of equipment, such equipment shall be furnished with standby equipment when necessary.
- (c) The equipment shall be designed to cope with any changes in the quality and quantity of seawater for the seawater electrolysis chlorine injection equipment. In addition, the equipment shall be designed to permit start-up through normal operation even after a long period of shutdown.
- (d) The equipment shall be designed so that the entire equipment can be tripped in sequence in the event of abnormal decrease in the seawater volume and/or abnormal rise of electrolysis current.
- (e) The system for the on site generation and distribution of sodium hypochlorite solution shall be supplied to provide continuous and intermittent shock chlorine dosing to the seawater intake to prevent biological fouling of heat exchangers surface and piping system.

The system shall be common, consisting of two (2) trains, and each train shall be designed to produce 100% duty.

The system shall be capable of continuously maintaining a total combined chlorine residual of 0.25 ppm minimum in the cooling water discharge from the main condensers and the closed cooling heat exchangers. The design continuous demand, **shall refer to clause 5 Performance Test and Guarantees item 5.3 Guarantee**. The Bidder shall verify that this is the correct chlorine demand.

The system shall also be capable of intermittent shock dosing of one unit by parallel operation of two (2) trains.

The method of hypochlorite generation shall be by the electrolysis of seawater.

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The cells shall be designed to operate without blockage and with minimal routine cleaning. All equipment necessary for any such routine automatic cleaning shall be provided by the Bidder.

Each train shall include one (1) duty electrolytic cell unit with one (1) duty transformer and rectifier.

If dilution water is required to maintain proper flow through the diffusers, the Bidder shall provide all piping (include waste water piping), instruments and controls for this function. The flow regulating valve shall be provided with isolating and bypass valves for maintenance and shock dosing purposes.

All personnel safety equipment shall be provided especially with the consideration that with the malfunction of equipment chlorine gas may be generated.

The Power supply shall be fed from 6.3 kV Switchgear or 400 V Power Distribution center to be installed by the Bidder.

4. DETAILED REQUIREMENTS

4.1 General

The equipment shall be constructed in a manner and of materials which have been well proven in similar plants and locations.

While both hydrogen separation and sodium hypochlorite solution tanks are specified, it may be possible to carry out both hydrogen removal and sodium hypochlorite solution storage in one (1) tank.

The Bidder may propose one (1) tank if it can clearly be shown that both specified storage and effective hydrogen removal will be achieved.

4.2 Electrolytic Cells

The hypochlorite generator electrolytic cell(s) is consisting of parallel cell assemblies (module). Each module shall have isolation valves and DC circuit breakers so that each module can be operated independently.

The generator shall have a design operating pressure of at least 10% higher than its operating pressure and will be able to withstand test pressure of 1.5 times its design pressure for 15 minutes.

The hypochlorite generator electrolytic cell(s) shall consist of plates/tubes to be connected at each end. The outer surface of plate/tubes shall be coated with a platinum alloy to give a service life guaranteed for at least five (5) years continuous operation. The Bidder shall advise the Owner of base metal and coatings recommended for this service and life.

Each cell shall be accurately spaced and contained by inert insulating supports. The cell(s) shall be fitted with external, electrical cathode and anode connectors. The generator modules shall be protected against blockage or low flow conditions by flow monitoring devices which shall isolate the power supply to the generator by activating DC circuit breakers or the AC main input and shall warn the operator.

All electrical connections shall be made externally to the outer surface of the generating cell. Through-the-wall electrical connection is not acceptable.

Hardware such as bolts, nuts, flat washers and lock washers used on the electrolytic cells shall be type 316 stainless steel.

4.3 Safety Features

4.3.1 Open Top Hydrogen Removal Tank

The sodium hypochlorite hydrogen removal tank shall be designed for the safe and efficient removal of hydrogen and be a cylindrical, vertical and open tank and shall be constructed of fiber reinforced plastic with a minimum 2.5 mm corrosion barrier of vinyl ester resin or the Owner approved equivalent. The top of tank shall be protected from debris and animal (e.g. bird) entry with a fiber reinforced plastic grating and provided with a fresh water foam breaking sprayer.

4.3.2 Lightning Arrester

Single mast type (free standing pole) lightning arrester with air terminal shall be provided.

4.3.3 Fence

Wire mesh of 2 m height, approximately 8 m apart from around the open top tank shall be provided to prevent access of unauthorized personnel.

4.3.4 Warning Sign Boards

The Bidder shall provide warning sign boards on the following :

- Building Fence and Ladder
- Chlorination Building
- Tank Ladder

4.4 Miscellaneous Requirements

All components, piping and valves shall be constructed of materials suitable for their chemical usage.

To the maximum extent feasible, the arrangement of all parts shall be such that any part, and especially those parts requiring replacement in the normal course of preventive maintenance, may be removed and replaced, either without disturbing adjacent parts or with a minimum amount of disturbance thereto.

All materials not specifically covered herein shall be of the best quality for the service contemplated.

Safety guards for all revolving shafts and shaft coupling and for other exposed moving parts shall be provided.

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter, and shall be given one prime coat of paint, suitable for the specified operating and environmental conditions. No paint having an asphaltic base shall be used.

The Bidder shall ship a sufficient quantity of paint for touching up, where prime coat has been damaged during shipment and erection. Paint shall be the same kind and quality as used for each piece of equipment in the shop.

Permissible noise levels shall be as specified in Part 4, Subsection 2.9.

Capacity of the tank shall be in accordance with manufacturer's recommendation.

4.5 Chlorination Facility

Chlorination Facility (Sodium hypochlorite solution injection) shall be provided for preventing the adherence of bacteria in desalination plant.

Chlorination Facility shall include of the followings.

Sodium hypochlorite solution Storage tank of more than seven (7) days capacity for injection shall be provided.

One (1) set of Sodium hypochlorite solution injection pump of 100% capacity (Two (2) Sea water Feed pump). The pump discharge shall be of the automatic control type and Sodium hypochlorite solution is injected into the sea water near the desalination sea water feed pump.

4.6 Chemical Injection Facility

The chemical injection facility shall be provided for preventing the adherence of scale, bacteria in the membrane element.

Storage solution tank of more than seven (7) days capacity for each injection facility shall be provided.

Two (2) sets of chemical injection pump of 100% capacity for each kind of chemical shall be installed, such as sodium by sulfite. The pump discharge shall be of the automatic control type. The liquid side of each pump shall be constructed of corrosion proof material.

Chemical injection pumps and accessories shall be per Part 4, Specification M-009. The piping, valves, etc shall be provided from solution tank to injection point.

The drainage piping from chemical tanks area shall be led to the desalination drainage pit, and the drainage water shall be sent to the waste water pond through the sump pump.

All drainage piping shall be of corrosion resistant material.

5. PERFORMANCE TESTING AND GUARANTEES

The Bidder shall conduct performance testing in accordance with Part 4, Subsection 2.6 (Performance Testing, Guarantee and First Inspection) shall guarantee the following system guarantees :

5.1 Site Test

The piping will be subject to a field hydrostatic leak test of 1.5 times of the design pressure after it is installed. All workmanship and materials shall be of

such quality as to ensure that no defects due to faulty fabrication or faulty materials will develop during the testing.

Site test shall be conducted by the Bidder to prove the guaranteed performance.

5.2 Shop Test

- (a) All pumps shall be hydrostatically tested.
- (b) Hydrostatic testing to at least 1.5 the maximum allowable working pressure prior to application of protective lining (if applicable).

5.3 Guarantees

The system shall be capable of continuously maintaining a total combined chlorine residual of 0.25 ppm in the cooling water discharge from the main condenser and shall be rated for an injection quantity of not less than 1.75 ppm equivalent chlorine into the intake of the C.W. pump.

In addition the system shall have the capability for shock treatment of an individual unit of the plant. This treatment shall allow, by parallel operation of common and unit trains, for an injection of up to 3.5 ppm of equivalent chlorine continuously.

The system shall be arranged and designed to provide complete flexibility of operation in terms of solution strength, dosing quantity and treatment timings.

6. QUALITY ASSURANCE / QUALITY CONTROL

6.1 General

Quality control shall conform to the requirements of part 4, subsection 2.7.

The quality control procedures specified, shall include but not be limited to the following :

- Welding
- Nondestructive testing
- Heat treating
- Packaging and Shipping
- Shop test procedures
- Cleaning and painting

The quality control documentation as specified in Part 4, Subsection 2.7 shall include, but not be limited to the following :

- Shop test reports
- Material certifications
- NDE reports
- All others documentation required by applicable codes and standards

6.2 Shop Tests for Pressure Vessels

All pressure vessels shall be subject to the following shop tests performed by the Bidder :

- (a) Magnetic particle testing of all sheared and burned edges and weld preps (ASTM E709)
- (b) Radiography of pressure boundary welds in accordance with ASME section VIII Division 1.
- (c) Spark testing at 15,000 V (if lined).

7. INSTRUMENTATION AND CONTROLS

7.1 General

- (a) The Bidder shall be responsible for the design, supply and installation of complete instrumentation and control systems and devices for the Chlorination Plant including, but not be limited to, those listed and described in this Technical Specification.
- (b) **The Bidder shall provide programmable logic controller (PLC) interface with DCS**
- (c) The Bidder shall also furnish and install serial link interface with the plant monitoring DCS.
- (d) Refer to Part 4, Technical Specification I-001 for design requirements for the instrumentation and control devices.
- (e) Refer to Part 4, Technical Specifications E-001 and E-230 for design requirements for the junction boxes, cables and wiring.

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7.2 Local Control Panel

The Bidder shall furnish and install a local control panel for the Chlorination Plant.

The local control panel shall contain all control and monitoring devices which do not need to be installed at local measuring or control location, and all operator interface such as indicators, push buttons, lights, control switches and alarms.

The local control shall also contain all necessary control system hardware/modules.

Refer to Part 4, Technical Specification I-002 for the design requirements for the local control panels.

The control panel shall include, but not be limited to, the followings :

- (a) One (1) back lighted window type annunciator in accordance with Technical Specification I-002 Local Control Panels to alarm the following as a minimum :
- Motor overload
 - Chlorine leak (if acid cleaning system is required)
 - Hydrogen leak
 - Rectifier power failure
 - High cell temperature
 - High reactor temperature
 - High rectifier temperature
 - Rectifier overload
 - Hydrogen dilution blower failure
 - Low hypochlorite concentration
 - Low and high hypochlorite tank level
 - Low and high hydrogen separation tank level (if applicable)
 - Loss of control power
 - Reduced air flow at rectifier enclosure
 - High differential pressure across a sea water strainer
 - High and low residual chlorine at condenser CW outlet

- (b) The annunciator shall be furnished with contacts to signal “Chlorination Plant Trouble” to the Plant Monitoring System when any one (1) of the above alarms has been initiated. The contacts shall open and reclose if a further alarm occurs before the previous one.
- (c) Control switches and status lights for all pump motors and for motorized valves.
- (d) Sodium hypochlorite generator flow indicator.
- (e) All hardware and interconnecting wiring to provide interlocking and automatic shut-down of all chlorination injection pumps if the level of the sodium hypochlorite storage tank becomes high.
- (f) Automatic start-up of the standby hydrogen dilution blower or failure of the running one. The discharge valve of the failed blower shall be automatically closed and the discharge valve of the standby blower shall be automatically opened.
- (g) Automatic shutdown of all sodium hypochlorite injection pumps if the level of hydrogen separation tank becomes low.
- (h) Automatic shutdown of all chlorination booster pumps if the level of the sodium hypochlorite storage tank becomes high.

7.3 Locally Mounted Devices

The Bidder shall furnish and install locally mounted devices for the Chlorination Plant.

One gaseous hydrogen and chlorine leak detectors to give an alarm on the local annunciator and a signal for the plant computer in the event of a hydrogen chlorine leak.

A high differential pressure alarm switch for the seawater strainer.

A level controller for the hydrogen separation tank. This controller shall modulate an air operated control valve downstream of the sodium hypochlorite injection pumps.

Pressure indicators with chemical seals at the discharge of each pump, on the suction header to each group of sodium hypochlorite injection and chlorination booster pump (Chemical seals are not required for air service).

A flow element and electronic type transmitter to measure the sodium hypochlorite effluent flow from the generating system.

7.4 Sampling Pump for Residual Chlorine Analyzer System

The Bidder shall furnish and install residual chlorine analyzer system with sampling pump, if necessary, for the Chlorination Plant.

One (1) amperometric type indicating residual chlorine analyzer complete with high and low residual chlorine alarm contacts for the Chlorine Plant control panel annunciator and the plant monitoring system.

A rotameter and throttle valve to bypass part of the sample flow from the pump discharge to drain.

A flow switch which shall stop the pump on loss of suction flow. Pump shall be of self-priming type.

Pump suction filter and discharge pressure indicator.

A manual flow regulating valve to preset the flow of sodium hypochlorite to the CW intake for normal dosing. The flow regulating valve shall be provide with isolating and bypass valves for maintenance and shock dosing purposes.

Pressure indicators with chemical seals upstream and downstream of the flow regulating valve and at the discharge of each pump.

A high differential pressure alarm switch for the seawater strainer.

Start and stop of motors at Chlorination Plant shall be made possible by touch system on Common Control Desk located at Central Control Room.

8. ELECTRICAL SPECIFICATIONS

8.1 General

The Bidder shall furnish and install all electrical equipment and accessories required for the Chlorination Plant.

Those equipment shall include but not be limited to the followings :

1. Motors.
2. Cables and Wiring, conduits, and cable trays.

3. Transformers, rectifiers and power supply and receiving facilities.
4. All electrical control, monitoring and instrumentation devices and other appurtenances required for this Specification M-407: Chlorination Plant shall be provided by the Bidder.

For general electrical requirements, see Part 4, Technical Specification, E-001.

8.2 Motors

For detailed electrical requirements, see Part 4, Technical Specification E-210 Low Voltage Motors.

8.3 Cables and Wiring, Conduits, and Cable Trays

Any cables and/or bus ducts required between transformer- rectifiers and electrolytic cells shall be furnished and installed by the Bidder.

For detailed electrical requirements see Part 4, Technical Specifications, E-230, E-231 and E-232.

8.4 Rectifiers

The rectifiers shall be provided with appropriate protective devices to detect and shut down individual electrolytic cell line for plugged cells or short circuited cells.

The rectifier enclosure shall be free standing and self supporting. It shall be designed for indoor operation with a degree of protection of IP 54 according to IEC 60529. All doors shall be gasketed.

If natural cooling is inadequate, the equipment shall be cooled at a maximum ambient temperature of 40°C by a dual forced air cooling system in such a manner that the equipment may be operated at full rated output power with one (1) cooling system inoperative. Reusable air filters which may be easily removed shall be provided.

A ground pad with compression type lugs for the ground cable shall be furnished near the bottom of the enclosure in accordance with Part 4, Technical Specification E-241.

Lifting eyes shall be provided for the rectifier cubicle, together with sufficient additional permanent or temporary removable bracing to permit lifting, skidding or rolling each assembly without damage.

To the maximum extent possible, the arrangement of parts shall be such that any part, and especially those parts requiring replacement in the normal course of preventive maintenance, may be removed and replaced, either without disturbing adjacent parts or with a minimum amount of disturbance thereto.

Factory tests of the rectifiers shall be in accordance with IEC-60146.

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